Kyle Sabado

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github.com/Kylx28 | Portfolio: kylx28.github.io/

Education

University of Toronto Sept. 2022 – May 2028
Bachelor of Applied Science in Computer Engineering, Minor in Robotics *Toronto, ON*

Relevant Courses: Matrix Algebra & Optimization, Probabilistic Reasoning, Real Analysis, Data Structures & Algorithms, Operating Systems, Computer Networks, Deep Learning, Control Systems, Robot Modeling & Control

Technical Skills

Languages: Python, C++, C, MATLAB, Java, Bash, Verilog
ML & Distributed Systems: PyTorch, PyTorch Profiler, vLLM, NCCL, Ray, Hugging Face, Weights & Biases
Tools & Frameworks: Linux, Git, Docker, ROS, OpenCV, Google Cloud Platform, GitLab

Experience

Reinforcement Learning Research Intern May 2026 – Present
MARMoT Lab (National University of Singapore) *Singapore*

- Developing a **PPO**-based **reinforcement learning** policy with decaying expert action priors to improve planner trajectories through shorter paths and motion optimization.
- Built a timed roadmap planning pipeline using **NVIDIA SWAGGER** for waypoint generation and Safe Interval Path Planning + LNS for generating **multi-agent pathfinding** trajectories.

AI Networking Intern May 2025 – April 2026
Huawei Canada *Toronto, ON*

- Improved **Mixture-of-Experts (MoE) LLM inference latency by 20%** through designing a topology-aware expert placement system in **Python**, using expert sharding and replication to reduce communication overhead.
- Developed a distributed **collective communication scheduler** for multi-tenant GPU workloads using **Python** and TCP/RPC, improving training throughput by up to **30%** under high contention scenarios.
- Trained and profiled LLMs across **16 GPUs** using **Megatron-LM**, **PyTorch FSDP**, and **PyTorch Profiler**, evaluating tensor, expert, and data parallelism to identify performance bottlenecks.
- Implemented **AlltoAllv** using **NCCL** primitives in **vLLM**, enabling variable-size data transfers for distributed MoE inference.

6G Networks Research Student May 2024 – Sept. 2024
WIRLab (University of Toronto) *Toronto, ON*

- Developed and simulated **EKF**-based cooperative localization algorithms in **MATLAB** for RIS-enabled wireless networks, achieving **centimeter-level** tracking accuracy.

Design Team & Projects

Drone Racing Team Member Sept. 2023 – Dec. 2025
University of Toronto Autonomous Drone Racing (UTADR) *Toronto, ON*

- Developed real-time **C++/ROS** localization pipelines using IMU-camera fusion and **FAST/ORB** feature tracking, enabling visual-inertial state estimation.

Mapping and Pathfinding Application | *C++* Sept. 2023 - May 2024

- Collaborated in a team of 3 to build a full-stack mapping application in **C++** using the **OpenStreetMap API**, enabling interactive navigation and route visualization.
- Implemented **A*** for efficient single-destination pathfinding and a custom **greedy heuristic** for traveling salesman-like optimization, reducing route computation time.

Publications

M. Ammous*, **K. Sabado***, M. Saif and S. Valaee, ”Cooperative Localization and Tracking Using RISs and Sidelink Communications,” 2025 IEEE Wireless Communications and Networking Conference (WCNC), Milan, Italy, 2025.